

Abdulaziz Alhaidan

AI Engineer

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SUMMARY

AI Engineer who builds and ships production LLM and computer-vision systems end-to-end — from data preparation and training through evaluation to REST-API deployment. Won 1st place at the Himmah Digital Camps hackathon for a multi-agent advisory system built with LangChain and FAISS, and independently designed, built, and launched an Arabic-first review platform now live in production. Focused on RAG pipelines, multi-agent orchestration, embeddings, and vector databases, with production integrations of OpenAI, Azure OpenAI, and Anthropic Claude.

TECHNICAL SKILLS

Languages: Python (primary), JavaScript, SQL

Generative AI & LLMs: LLM integration (OpenAI, Azure OpenAI, Anthropic Claude), multi-agent systems & orchestration, RAG (built from scratch), embeddings, vector databases (FAISS, ChromaDB), Transformers & BERT, Hugging Face, fine-tuning, prompt engineering (prompt chaining, structured extraction)

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, CNNs & transfer learning (ResNet50), RNN/LSTM, time-series forecasting, regression, decision trees, random forests, ensembles, K-Means, KNN

Evaluation & Data: EDA (pandas), feature engineering, hyperparameter tuning, cross-validation, class-imbalance handling, confusion matrices, precision/recall, failure-case analysis

MLOps & Deployment: MLflow experiment tracking, ONNX export, model serialization, FastAPI model serving, Docker, CI/CD fundamentals, cloud deployment (Hugging Face Spaces, Railway, Render, Vercel)

Backend & Cloud: FastAPI, REST APIs, Azure (AI Foundry, Azure OpenAI, AI Services), AWS (IAM, EC2, S3), Railway

Frontend & Practices: React.js, Next.js, Git, Responsible AI

SELECTED PROJECTS

AI Board Room — Multi-Agent Executive Advisory System | 1st Place, Himmah Digital Camps · Jun 2026 | ai-board-room-2wtb.onrender.com

- Designed the multi-agent architecture and consensus flow, defining each agent's role boundaries, retrieval scope, and how conflicting recommendations resolve into a single synthesized output.
- Built a multi-agent system simulating a corporate advisory board: specialized CFO, Legal Advisor, and Market Strategist agents deliberate and vote, with a Chairman agent synthesizing the final consensus recommendation.
- Engineered a FAISS-backed RAG layer over company documents so each agent grounds its reasoning in retrieved, role-specific context.
- Orchestrated agent workflows and tool calls with LangChain; served the system through a FastAPI backend and React interface, deployed and publicly accessible.

Tech: Python, LangChain, FAISS, FastAPI, React

AgroCure — Plant Disease Detection & Advisory System | Bootcamp Capstone · Jun 2026 | abdulazizcsdev-agrocure-app.hf.space

- Trained a ResNet50 transfer-learning classifier across 36+ disease classes on ~7,300 images, using 5-fold cross-validation and a WeightedRandomSampler to correct class imbalance.
- Added a retrieval (RAG) layer for rare classes and fine-tuned a new class on an 80-image dataset collected and labeled in-house.
- Deployed the trained model as a public inference service on Hugging Face Spaces, wrapped in a FastAPI + Docker serving layer; tracked experiments and model selection with MLflow on DagsHub.
- Drove the technical plan across data pipeline, training, and deployment, and guided a four-person team through model selection and evaluation decisions.

Tech: Python, PyTorch, ResNet50, FAISS, MLflow, Docker

Hanak — Arabic Restaurant Review Platform | Jul – Aug 2026 | hanak-rosy.vercel.app

- Designed and shipped a full-stack Arabic-first review platform end-to-end — product architecture, database schema, UI, and deployment — from concept to live release.
- Built a Postgres data model on Supabase with trigram (pg_trgm) fuzzy search over Arabic venue names, and enforced row-level access with SECURITY DEFINER policies for admin operations.
- Solved the duplicate-venue problem by modeling each restaurant's menu as the canonical unit, keeping reviews consistently attributed across branches and name variants.

Tech: Next.js, Supabase (Postgres), Vercel

Aime — Arabic-First AI Voice Assistant | Apr 2026

- Built and deployed an Arabic-first voice assistant integrating OpenAI Whisper (STT), Anthropic Claude (reasoning), and ElevenLabs (TTS with voice cloning).
- Implemented a LangChain + ChromaDB RAG pipeline for grounded, context-aware responses; served via FastAPI and deployed on Railway.
- Handled Arabic-specific challenges across the pipeline — dialect variation in transcription and diacritic-insensitive retrieval — to keep responses grounded in the correct source context.

Tech: Python, FastAPI, LangChain, ChromaDB, Whisper, Claude API

Between the Lines — AI X-Archive Analyzer | May 2026 | x-tweets-analyzer.vercel.app

- Built an archive analyzer that surfaces engagement patterns and generates content insights with the Claude API, using client-side ZIP parsing so user data never leaves the browser.
- Delivered interactive dashboards with heatmaps, trend charts, and weekly content plans.

Tech: Next.js, FastAPI, Claude API

Additional Projects

- Weather-Driven Car Wash Demand Forecasting (2025) — Built a Scikit-learn model forecasting daily customer demand from weather signals, then shipped it as a FastAPI REST service integrated into an existing car wash management platform to inform staffing and inventory planning. *Self-initiated extension.*
- Car Wash Management Platform (2025) — *Graduation Project* — Full-stack booking, order-tracking, and payment system with role-based REST APIs for mobile and admin clients.
- Online Multiplayer Party Game (2025) — Real-time architecture with React and Socket.IO, handling concurrent state synchronization across users.

EXPERIENCE

Cooperative Training, Quality Assurance — T2, Riyadh | Feb – May 2025 · University co-op requirement

- Designed and executed functional and regression test suites across releases; tracked defects to resolution and validated fixes before sign-off.
- Built a systematic approach to failure-case analysis and edge-case discovery — the same discipline now applied to evaluating model outputs and RAG retrieval quality.

EDUCATION & TRAINING

AI Engineering Bootcamp — Saudi Digital Academy (SDA) | May – Jul 2026

- Intensive, project-based program covering the full ML engineering lifecycle, from classical ML to production LLM systems.
- Fine-tuned transformer models (BERT) with Hugging Face for sentiment analysis and text classification; implemented RNN/LSTM, Seq2Seq, and time-series forecasting architectures.
- Applied advanced prompt engineering (prompt chaining, DSPy) for structured extraction from unstructured text; packaged models for production via ONNX export and serialization.

B.Sc. in Computer Science — Majmaah University | 2020 – 2025 | GPA 4.01 / 5

CERTIFICATIONS & COURSES

- IBM Data Science Professional Certificate — 2025
- DeepLearning.AI TensorFlow Developer Specialization — 2025
- AWS Cloud Practitioner Essentials (course) — 2025

LANGUAGES

Arabic (Native) · English (Advanced)